



Understanding Cloud Storage Mechanisms - Explain how data is stored, managed, and accessed in the cloud.

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Objective

- ❖ Understand the concept of cloud storage.
- ❖ Explain how data is stored in the cloud.
- ❖ Learn how cloud data is managed.
- ❖ Describe how users access cloud storage.
- ❖ Identify the advantages and applications of cloud storage.



Core Technical Explanation

- **Distributed Storage:** Data is stored across multiple servers in different data centers.
- **Virtualization:** Physical storage resources are combined and managed as virtual storage.
- **Data Replication:** Multiple copies of data are maintained to ensure reliability and fault tolerance.
- **Secure Data Access:** Data is protected using encryption, authentication, and secure network protocols.
- **Scalability & High Availability:** Storage capacity can be expanded on demand while ensuring continuous access to data.

Architecture Diagram

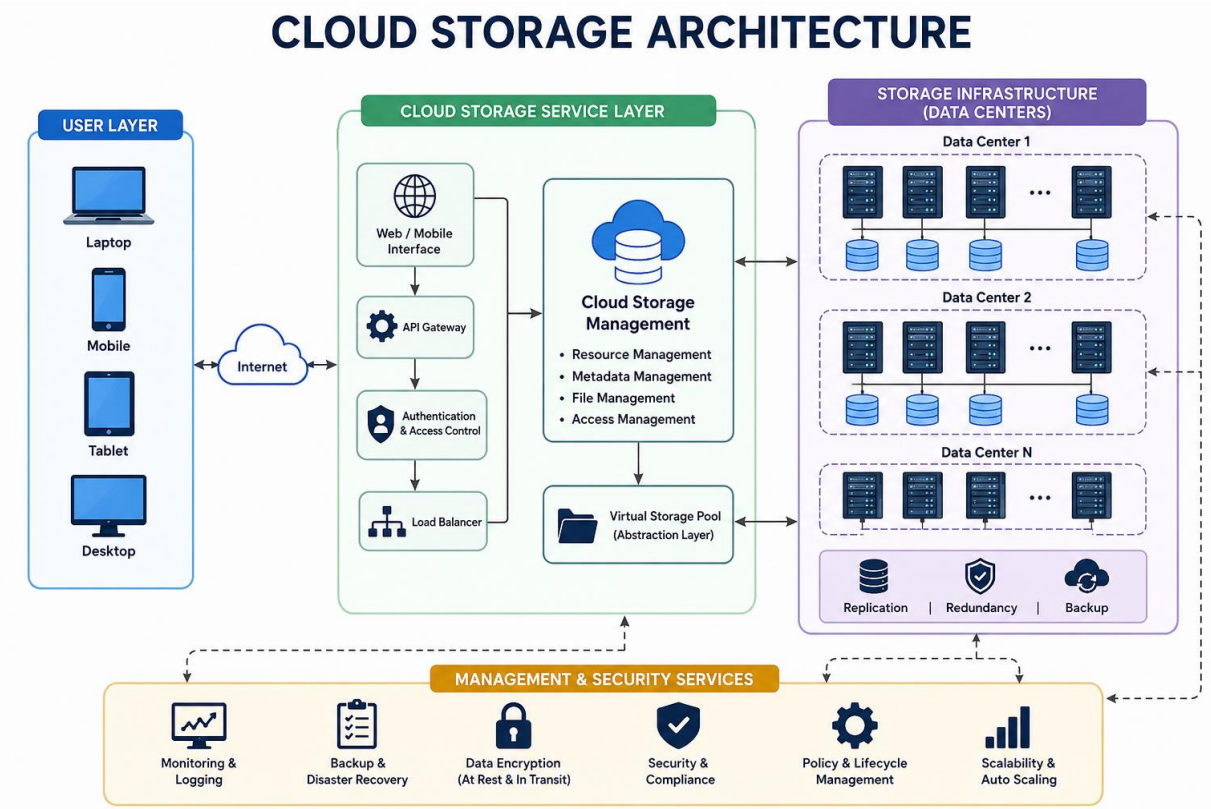
User Layer: Users access cloud storage through devices connected to the internet.

Cloud Service Layer: Processes user requests and manages authentication and storage services.

Virtual Storage Layer: Combines physical storage into a single virtual storage pool.

Storage Infrastructure: Stores data across distributed data centers with replication and backup.

Management & Security: Protects data using encryption, monitoring, and automatic scaling.



Real Life Cases

Students: Store assignments, presentations, and study materials and access them from any device.

Businesses: Share files and collaborate with teams in real time using cloud platforms.

Healthcare – Securely store and access electronic patient records with appropriate security and compliance controls.

Media & Entertainment: Store and manage large collections of photos, videos, and digital content.

Backup & Disaster Recovery: Automatically back up important data so it can be restored after system failures or other disruptions.

Conclusion

- Cloud storage is a modern technology that enables secure, scalable, and reliable data storage over the internet. It improves accessibility, reduces infrastructure costs, and ensures data protection through backup, replication, and encryption.

Key Takeaways

- Cloud storage allows data access from anywhere using the internet.
- Data is protected through encryption, backup, and replication.
- Virtualization improves storage efficiency and resource utilization.
- Cloud services provide scalability and high availability.
- Cloud storage is widely used in education, healthcare, business, and entertainment.

References

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